

SSRN WORKING PAPER

AI Presence Drift: A Longitudinal Re-Baseline of Five Brand Categories

One-Week Stability Validates the AIAS Measurement Program; Pattern 1 Cross-Model Spread, Pattern 4 Discourse-Language Bias, and Pattern 6 Phantom-Brand Persistence All Replicate

Working Paper · Version 0.9 · Longitudinal Re-Baseline (Five Categories)

Pablo Ulpiano González Castro

School of Visual Arts, MPS Branding Program, New York, NY
(primary academic affiliation)

Third System[™] (research entity; data archive and methodology venue)

Correspondence: pablou@pablou.com · pablou.com

ORCID: <https://orcid.org/0009-0003-8968-9990>

8 May 2026

Working paper. Not under peer review. Pre-registered.

Abstract

This paper reports a longitudinal re-baseline of the five categories first measured in the v0.6 cross-category baseline of the AIAS measurement program — project management software, premium running shoes, premium olive oil, premium facial skincare, and personal finance applications — at a second time point approximately seven days later. The study closes Phase 2 of the AIAS program by giving each baseline category a t_2 measurement and provides the data spine for Phase 3 construct-validity work. Six hypotheses with explicit numerical thresholds were committed to a pre-registration document locked at git commit f8cebbd (tag v0.9-prereg-locked) prior to any t_2 data collection. Methodology decisions — the matched 2-model subset (claude-sonnet-4-6 and gpt-5.4-mini, the original v0.6 lineup) for longitudinal deltas, retroactive mode classification of v0.6 data, and registry freeze at v0.6's final state — were locked at git commit e6428a4 prior to the pre-registration.

The headline result is decisive. Five formal hypotheses were confirmed and the sixth landed at the predicted stability band. Per-brand drift across all 5 categories on the matched subset ($n = 103$ brand-level deltas) was small: 88.3 percent of brands within ± 5 percentage points and 99.0 percent within ± 10 percentage points (H1 confirmed). Top-three brands at t_1 remained in the top-five at t_2 in every category (H2 confirmed, 5 of 5). Within-category brand-presence variance ordering across the five categories at t_2 correlated with t_1 at Spearman $\rho = 0.8$ (H3 confirmed). The Mint phantom-brand stayed at 41.7 percent matched-subset gross Presence at t_2 versus 44.8 percent at t_1 — a -3.1 percentage-point delta within the predicted stability band (H4 stability, the predicted outcome). Pattern 4 discourse-language bias replicated under the strict reading: Spanish olive oil aggregate Presence at 11.5 percentage points fell below the 12.5-percentage-point threshold derived from the International Olive Council production share, and K-beauty aggregate Presence in skincare at 1.0 percentage point fell well below the 5-percentage-point threshold (H5 confirmed strict; partially confirmed under an inclusive reading that adds Graza, a US-headquartered Spanish-sourced brand). Per-brand cross-model spread between the two matched-subset models correlated between t_1 and t_2 at Pearson $r \geq 0.7$ in every category (H6 confirmed, 5 of 5).

A post-hoc test of v0.6's Pattern 1 (cross-model variance and discourse coherence) at the category level — distinct from H3's within-category brand variance test — was performed against the same matched-subset data. Per-category mean cross-model spread between Sonnet and gpt-5.4-mini ranks at Spearman $\rho = 0.800$ between t_1 and t_2 , and at $\rho = 0.900$ between t_2 and v0.6's qualitative ordering (personal finance > olive oil > project management > skincare > running). Pattern 1 as v0.6 stated it replicates at v0.9 with the t_2 ranking better-matched to v0.6's narrative than t_1 is.

A meta-finding emerged from the H5 work: the v0.6 skincare registry contains a single K-beauty brand among thirty-one entries, mirroring inside the registry construction the same Anglo-discourse bias the test

is designed to detect. The H5 strict-versus-inclusive sensitivity confirms what v0.6 §4.4 already articulated — that discourse-language bias operates at the brand-marketing-language tier rather than country-of-origin — by isolating the diagnostic case of Graza, a Spanish-sourced English-marketed brand whose inclusion in the Spanish cohort breaches the H5 threshold while its exclusion confirms it.

The combined result set is interpreted as a longitudinal validity argument for the AIAS measurement program. AI Presence behaves like a real underlying construct — drift is small, leaderboards are stable, structural variance patterns replicate (both within-category brand variance and cross-model spread by discourse coherence), the phantom-brand persistence mechanism holds across waves at gross Presence, and cross-model relative behavior is preserved. The argument unblocks Phase 3 construct-validity work, which requires at least three categories at two time points and now has data for five.

Keywords

AI brand visibility; longitudinal validity; pre-registered measurement; brand presence; large language models; drift; phantom-brand persistence; discourse-language bias; cross-model variance; AIAS

JEL Classification

M31 (Marketing) — primary; L86 (Information and Internet Services; Computer Software); C81 (Methodology for Collecting, Estimating, and Organizing Microeconomic Data); D83 (Search; Learning; Information and Knowledge; Communication; Belief; Unawareness)

Paper status

Working paper. Not under peer review. Pre-registered longitudinal re-baseline closing Phase 2 of the AIAS measurement program. Builds directly on the cross-category baseline reported in González Castro (2026), “AI Presence Measurement Across Consumer Categories” (SSRN ID 6720959). Replicates Pattern 1 (cross-model variance and discourse coherence), Pattern 4 (discourse-language bias) refined hypothesis as stated in v0.6 §4.4, and Pattern 6 (phantom-brand surfacing). Engages the recommendation-slot persistence reframe of phantom-brand surfacing developed in González Castro (2026), “A Designed-for-Test Measurement of Phantom-Brand Presence in Large Language Model Outputs” (SSRN ID 6721779) §3.2, while measuring the gross-Presence variable common to both v0.6 and v0.7. Replicates and operationalizes Pattern 4 (refined hypothesis) reported in González Castro (2026), “A Designed-for-Test Measurement of Discourse-Language Bias in Large Language Model Brand Recommendations” (SSRN ID 6728000). Companion empirical artifact to González Castro (2026), “AI Availability: Extending Mental and

Physical Availability into Algorithmic Retrieval” (SSRN ID 6659000). Methodological reference: González Castro (2026), “AIAS Presence Measurement Protocol v1.1” (SSRN ID 6722319). Pre-registration document (PRE_REGISTRATION_v09_rebaseline_v1.0.md, locked at git commit f8cebbd, tag v0.9-prereg-locked) and underlying datasets released alongside this paper at [osf.io/\[id-pending\]/](https://osf.io/[id-pending]/).

1. Introduction

The cross-category baseline reported as v0.6 of the AIAS measurement program (González Castro 2026, “AI Presence Measurement Across Consumer Categories”) established a snapshot of brand visibility in large language model (LLM) outputs across five consumer categories at a single time point in late April 2026. The baseline produced six structural patterns — including cross-model variance and discourse coherence (Pattern 1), comparison-versus-discovery asymmetry (Pattern 2), divergence from consumer awareness (Pattern 3), discourse-language bias in cross-lingual categories (Pattern 4), default reinforcement and discourse-prompt alignment (Pattern 5), and phantom-brand surfacing in defunct brand cases (Pattern 6) — and a single-time-point dataset across approximately one hundred registered brands. Two designed-for-test follow-up studies (v0.7 phantom-brand persistence; v0.8 discourse-language bias) extended Patterns 4 and 6 to category-specific resolution and produced a theoretical reframe of phantom-brand surfacing as recommendation-slot persistence rather than knowledge-freshness lag (González Castro 2026, v0.7 §3.2). The single-time-point baseline left open a question that no snapshot can answer: *does AI Presence behave like a real underlying construct, or like measurement noise?* Without longitudinal evidence, the v0.6 baseline patterns are observations whose stability is unknown, and the construct-validity work that the AIAS program targets in Phase 3 cannot be undertaken — Phase 3 requires at least three categories measured at two time points correlated against external consumer-tracking data.

The longitudinal re-baseline reported here addresses both gaps directly. The same five categories from v0.6 were re-measured at a second time point approximately seven days later, against a frozen registry, with the same six pre-registered prompts anchored to Category Entry Points. The matched 2-model subset for the longitudinal claims comprises the original v0.6 model lineup (Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini); the t_2 measurement adds four parallel-baseline models (Anthropic Opus 4.7, OpenAI gpt-5.5, Google Gemini 2.5 Flash, xAI Grok 4.1 Fast) for cross-version completeness, but these are reported as parallel new baselines outside the longitudinal frame. The methodological choice preserves clean t_1/t_2 attribution: longitudinal deltas are computed only on the subset of models that were measured at t_1 , so any divergence is attributable to AI behavior change rather than to model-set expansion.

Six hypotheses with explicit numerical thresholds were committed to a pre-registration document (PRE_REGISTRATION_v09_rebaseline_v1.0.md) locked at git commit f8cebba (tag v0.9-prereg-locked) prior to any t_2 data collection (Nosek et al. 2018). The hypotheses isolated specific candidate properties of the LLM tier: brand-level drift noise floor (H1); top-of-leaderboard stability (H2); within-category brand-presence variance preservation across categories (H3, related to but not identical with v0.6’s Pattern 1, which concerns cross-model spread; the latter is tested separately as a post-hoc replication in §3.7); phantom-brand persistence on Mint, the canonical phantom case from v0.6 §4.6 (H4, measured at gross

Presence per the variable v0.6 reported); replication of the v0.6 §4.4 refined hypothesis on discourse-language bias for Spanish olive oil and K-beauty skincare brands (H5); and cross-model relative-behavior stability between the two matched-subset models (H6). A seventh observation — mode-distribution stability after retroactive mode classification of v0.6 — was deferred to exploratory status because no prior calibration existed for what threshold counts as “meaningful shift” in mode share.

The headline result is decisive. Five formal hypotheses were confirmed and the sixth landed at the predicted stability band. Drift across the five categories on the matched subset was small (88.3 percent of brands within ± 5 percentage points). Leaderboards were stable across all five categories. Within-category brand-presence variance ordering preserved at Spearman $\rho = 0.8$. Mint persisted at 41.7 percent gross Presence at t_2 versus 44.8 percent at t_1 , within the predicted ± 5 -percentage-point stability band. Spanish olive oil under-surfacing and K-beauty under-representation in skincare both replicated. Cross-model relative behavior between Sonnet and gpt-5.4-mini correlated between t_1 and t_2 at Pearson $r \geq 0.7$ in every category. A post-hoc category-level cross-model spread test of v0.6 Pattern 1 — distinct from H3’s within-category variance test — produced Spearman $\rho = 0.800$ between t_1 and t_2 rankings and $\rho = 0.900$ between t_2 ranking and v0.6’s narrative ordering. The combined set supports interpreting AI Presence as a real, structured property of the LLM tier rather than as the residue of measurement noise — and unblocks the Phase 3 construct-validity program.

Two findings worth highlighting in the introduction emerged outside the formal pre-registration. *First, the H5 strict-versus-inclusive sensitivity confirms what v0.6 §4.4 already articulated.* The v0.6 paper observed that “Tatcha, a Japanese-themed skincare brand founded in San Francisco with American-English marketing, scored 18 percent Presence in skincare. Beauty of Joseon, a Korean brand with Korean-language primary marketing despite some US-targeted English content, scored zero. The cultural reference (Japanese aesthetic) does not appear to be the constraint. The marketing-discourse language does.” (González Castro 2026, v0.6 §4.4). The refined hypothesis from v0.6 was that LLMs under-surface brands whose primary marketing discourse is conducted in a language under-represented in AI training data, even when the cultural reference is well-known in English. The v0.9 H5 sensitivity check operationalizes that refined hypothesis with explicit numerical thresholds and surfaces a sharper diagnostic case — Graza, a US-headquartered, Spanish-sourced, English-marketed D2C brand at approximately 14 percent matched-subset Presence. Including Graza in the Spanish cohort moves the aggregate from 11.5 percentage points (below the threshold) to 14.2 percentage points (above the threshold). The 2.7-percentage-point swing isolates the brand-marketing-language tier from country-of-origin cleanly. The contribution of v0.9 H5 is therefore not the framing — that is v0.6’s — but its longitudinal replication, its operationalization with pre-registered numerical thresholds, and the Graza diagnostic.

Second, the K-beauty registry coverage is itself a finding. The v0.6 skincare registry contains thirty-one

brands, of which exactly one (Beauty of Joseon) is a Korean-headquartered K-beauty brand discoursed primarily in Korean-language category media. The under-representation in the registry construction parallels the discourse-language bias the test is designed to detect. The implication is that the registry-construction process inherits, at the analyst tier, the same Anglo-discourse coverage gap that produces the LLM-tier discourse-language bias. The observation is reported as a methodological reflexivity caveat in §6.

The remainder of the paper is organized as follows. Section 2 describes the pre-registered method, including the matched-subset framing for the longitudinal claim, registry freeze with documented metadata anomalies, retroactive mode classification of v0.6, and pre-registration. Section 3 reports the six findings in narrative form, plus a post-hoc Pattern 1 replication test (§3.7) and exploratory mode-distribution observations (§3.8). Section 4 reports pre-registered hypothesis outcomes in tabular form. Section 5 develops the longitudinal validity argument: drift, leaderboard stability, structural-pattern replication, and the phantom-brand persistence and discourse-language bias replications interpreted together. Section 6 addresses limitations including the single-pair longitudinal subset, the seven-day inter-measurement interval, the registry-construction reflexivity exposed by the K-beauty observation, and the gross-Presence-versus-naive-phantom-rate question that v0.7 unpacked but v0.9 does not. Section 7 outlines Phase 3 work.

2. Method

The measurement followed the AIAS Presence Measurement Protocol v1.1 (González Castro 2026), with the longitudinal-re-baseline addendum (matched-subset framing for the longitudinal claim; parallel new-baseline framing for the four added models) developed and locked in the v0.9 methodology decision note (docs/v09_DECISION_NOTE.md) at git commit e6428a4 prior to the pre-registration.

2.1 Longitudinal Frame and Matched Subset

The v0.6 measurement was conducted with two models per category: Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini. The Phase 2 measurement program (v0.7 phantom-brand and v0.8 discourse-language) expanded the lineup to six models across four labs. The v0.9 measurement runs the full Phase 2 lineup (six models) at t_2 , but for the longitudinal claim — the $t_1 \rightarrow t_2$ comparison that constitutes the substance of H1 through H6 — only the matched two-model subset (Sonnet 4.6 and gpt-5.4-mini) is used. The four additional models (Anthropic Opus 4.7, OpenAI gpt-5.5, Google Gemini 2.5 Flash, xAI Grok 4.1 Fast) report as parallel new baselines: a first measurement for each at t_2 , providing the t_1 reference for that model's own future longitudinal work. The framing preserves clean t_1/t_2 attribution: any $t_1 \rightarrow t_2$ delta on the matched subset is attributable to AI behavior change between waves, not to model-set expansion. The framing is documented in the v0.9 methodology decision note (Decision #1).

2.2 Registry and Prompt Freeze

The registry and prompt set were frozen at v0.6's final state per Decision #3 of the methodology note. The registries used at t_2 are the per-category JSON files committed to the repository as of the pre-Phase-2 reorganization (git commit 179b3d7), which split the original consolidated `brands.json` into per-category `registries/brands_<category>.json` files. The reorganization was structural — it added the per-category file layout but preserved the underlying brand lists and prompt text — and is treated for the purposes of the registry-freeze decision as a schema-only change rather than a content revision. Inspection of the v0.6 measurement CSVs confirmed that the brand lists and prompt text in the t_1 measurements match the brand lists and prompt text loaded at t_2 in all five categories.

One documented metadata anomaly affects the personal finance v0.6 measurement. The `brand_registry_version` field on the personal finance v0.6 rows reads `v2-skincare`, inherited from a module-level constant in the pre-Phase-2 runner that was not reset between the skincare and finance sessions on 30 April 2026 (skincare ran at 17:51 UTC; finance ran at 18:16 UTC, twenty-five minutes later). Inspection of the v0.6 finance raw responses confirmed that the prompts that fired were the finance prompts and the brand mentions extracted are personal finance brands; the metadata stamp is a cosmetic carryover, not a measurement defect. The anomaly is disclosed here for transparency and does not affect any H1 through H6 score.

2.3 Prompt Set

The same six prompts used at t_1 were used at t_2 , anchored to Category Entry Points per AIAS Protocol §3.1: FUNCTIONAL (p1), CONTEXTUAL (p2), CONSTRAINT (p3), IDENTITY (p4), DISCOVERY (p5), COMPARISON (p6). Per-category prompt text was re-used verbatim from v0.6.

2.4 Model Lineup at t_2

Six frontier models from four labs were measured at t_2 . Two of the six (Anthropic Sonnet 4.6 and OpenAI gpt-5.4-mini) are the matched-subset models — present in v0.6 and used for the longitudinal claim. Four (Anthropic Opus 4.7, OpenAI gpt-5.5, Google Gemini 2.5 Flash, xAI Grok 4.1 Fast) are parallel new baselines. The Phase 2 lineup matches v0.7 and v0.8, enabling cross-version comparison of the parallel-baseline measurements with the same model lineup observed in those studies.

2.5 Measurement Volume and Failed-Call Recovery

Each (prompt, model, run) tuple was measured once at temperature 0.7 where supported. The target measurement volume per category at t_2 is 6 prompts $\times$ 6 models $\times$ 8 runs = 288 cells, totalling 1,440 cells across

the five categories. The first-pass run on project management software produced 233 of 288 successful measurements (80.9 percent), driven by an OpenAI billing-quota ceiling that was reached partway through the run. After the billing ceiling was raised, the remaining four categories' first-pass runs produced 287 of 288 (running shoes), 287 of 288 (olive oil), 288 of 288 (skincare), and 282 of 288 (personal finance) — totalling 96.5 percent first-pass completion across the four categories that ran after the billing fix. Failed cells were recovered through the pre-registered failed-call recovery cycle (AIAS Protocol §4.3) per provider, yielding 1,440 of 1,440 successful measurements (100.0 percent) after recovery. One observation worth flagging: Google Gemini 2.5 Flash retries during the project management software recovery pass averaged 322 seconds per call (versus approximately 8 seconds during the original run), with all eight retried cells eventually returning successfully. The slow-response episode is bounded to that recovery pass and does not affect the per-cell measurement; it is disclosed as a methodological observation.

2.6 Mode Classification, Retroactive at t_1

Per Decision #2 of the methodology note, the mode classifier developed during v0.7 and v0.8 was applied retroactively to the v0.6 raw measurements as well as forward to the v0.9 raw measurements, so that t_1 and t_2 datasets carry mode classifications under the same scheme. The classifier assigns each successful measurement to one of five modes: brand, mixed, component, authority, refusal. Cross-lab AI audits (twenty-five-row stratified samples per (category, wave) file) were generated for inter-rater agreement work; the audits are deposited but not analyzed for v0.9 hypothesis scoring, which uses the brand-surfacing macro unit (per AIAS Protocol §6.5 reframe convention) at which 96 percent agreement was established in v0.8. Mode-distribution stability between t_1 and t_2 is reported as exploratory observation in §3.8.

2.7 Pre-Registration

Six hypotheses with explicit numerical thresholds were locked in PRE_REGISTRATION_v09_rebaseline_v1.0.md at git commit f8cebbd, with tag v0.9-prereg-locked, prior to any t_2 data collection. Pre-registered methodological risks (the finance v2-skincare metadata stamp; possible undisclosed model updates between t_1 and t_2 ; rate-limit and API-failure handling; parallel-baseline failure as not contaminating the longitudinal claim) were documented at lock time. Hypotheses, predicted thresholds, and outcomes are reported in Table 1 (§4).

3. Results

3.1 Finding 1 — Drift across categories is small and tightly bounded.

Across all 103 brand-level deltas computed on the matched subset (the union of registered brands across the five categories, excluding any brand not in the v0.6 final-state registry), 88.3 percent of brands surfaced at t_2 within ± 5 percentage points of their t_1 Presence, and 99.0 percent surfaced within ± 10 percentage points. The H1 confirmation thresholds were ≥ 70 percent within ± 5 pp and ≥ 90 percent within ± 10 pp, both cleared with margin. The empirical noise floor of AI Presence measurement on a one-week interval, on the matched subset, is therefore approximately ± 5 percentage points for the modal brand and approximately ± 10 percentage points for ninety-nine of every hundred. Among the largest movers, no single brand exceeded ± 20 percentage points; the distribution is concentrated near zero.

3.2 Finding 2 — Leaderboards are stable.

In each of the five categories, the top-three brands by matched-subset Presence at t_1 remained in the top-five at t_2 . The result confirmed at 5 of 5 categories with no exceptions, well above the 5-of-5 threshold for the headline confirmation band. The leaderboard stability is the most directly category-management-relevant of the six findings: brand managers and category buyers tracking AI Presence as a leading indicator can rely on the top-of-leaderboard composition across at least the seven-day measurement interval.

3.3 Finding 3 — Within-category brand-presence variance ordering preserves.

The pre-registered H3 measures whether the within-category dispersion of brand-presence rates — how spread out brands are within each category, computed as the standard deviation of per-brand Presence values — preserves its rank-ordering across categories between t_1 and t_2 . The Spearman rank correlation of within-category variance between t_1 and t_2 across the five categories is $\rho = 0.8$, above the 0.7 threshold for confirmation and well above the 0.4 threshold for partial confirmation. The within-category variance ordering is therefore stable across waves.

A clarification on what H3 tests and what it does not. H3's statistic is *within-category brand variance*: how much the registered brands within a single category differ from each other on Presence. This is related to but distinct from v0.6's Pattern 1 (Cross-Model Variance and Discourse Coherence), which is *between-model spread per brand within a category* — a different statistic computed at the brand $\times$ model level rather than at the category level. Pattern 1's claim is that fragmented-discourse categories (personal finance, olive oil, project management) produce wider cross-model spreads than coherent-discourse categories (running shoes, skincare). H3 does not directly test that claim. The cross-model spread test is reported separately in

§3.7 as a post-hoc replication.

3.4 Finding 4 — Mint persists at gross Presence.

The Mint phantom-brand observation from v0.6 §4.6 (the brand was Intuit’s leading personal finance application until its decommissioning in March 2024) holds at t_2 . Mint’s matched-subset gross Presence in personal finance was 44.8 percent at t_1 and 41.7 percent at t_2 , a delta of -3.1 percentage points within the predicted ± 5 -percentage-point stability band. The pre-registration explicitly named stability rather than decay as the predicted outcome under the corpus-aging mechanism articulated in v0.6 §4.6: one additional time point seven days later is not enough for a defunct brand to be displaced from the LLM corpus, particularly when the brand’s prior dominance generated extensive English-language discourse that persists in both training data and continuing inferential context. The H4 finding is therefore a confirmation of the predicted-stability outcome.

A note on what H4 measures versus what v0.7 unpacked. H4 measures *gross Presence* — the rate at which Mint surfaces in the model’s output regardless of how the model contextualizes it (with or without caveat about the brand’s decommissioning, with or without correction). This is the variable v0.6 §4.6 reported (Mint at 44 percent gross Presence). The v0.7 designed-for-test follow-up (González Castro 2026, “Phantom-Brand Presence”) reframed the phantom mechanism from knowledge-freshness lag to *recommendation-slot persistence*, distinguishing the *naive-phantom rate* (model presents the brand as fully live with no caveat — 1.7 percent for BBB in v0.7) from gross Presence (38.2 percent for BBB). The v0.9 H4 measurement does not unpack this distinction; the pre-registration tested the gross Presence variable common to v0.6, on which the longitudinal stability claim is supported. A v0.10 measurement could pre-register a naive-phantom-rate stability hypothesis on Mint specifically and would extend the v0.7 reframe into the longitudinal frame.

3.5 Finding 5 — Pattern 4 replicates at v0.9, with the v0.6 §4.4 refined hypothesis operationalized.

The v0.6 §4.4 baseline observed in two cross-lingual categories (premium olive oil and premium facial skincare) that brand-presence rates appeared to track marketing-discourse language rather than country of origin or cultural reference. The v0.6 paper’s diagnostic case for the refinement was the contrast between Tatcha (Japanese-aesthetic skincare brand founded in San Francisco with American-English marketing — 18 percent Presence) and Beauty of Joseon (Korean brand with Korean-language primary marketing — zero percent Presence): the cultural reference does not constrain Presence; the marketing-discourse language does. The v0.6 refined hypothesis stated “LLMs under-surface brands whose primary marketing discourse is conducted in a language under-represented in AI training data, even when the cultural reference is well-

known in English.” The v0.8 designed-for-test on premium kitchen knives (González Castro 2026, “Discourse-Language Bias”) confirmed the mechanism on a third category. The v0.9 H5 contribution is the operationalization of the refined hypothesis with explicit pre-registered numerical thresholds applied to the original two cross-lingual categories at t_2 .

The strict-reading H5 outcome confirms the pattern. Spanish olive oil aggregate Presence, restricted to the two registered brands that present themselves to consumers as Spanish-discourse-coverage (Castillo de Cavena and Núñez de Prado), measured 11.5 percentage points — below the 12.5-percent threshold derived from Spain’s approximately 50-percent share of global olive oil production (International Olive Council). K-beauty aggregate Presence in skincare, comprising the single registered K-beauty brand (Beauty of Joseon), measured 1.0 percentage point — well below the 5-percent threshold.

A sensitivity check produces a sharper instance of the v0.6 refined hypothesis. When Graza — a US-headquartered, Spanish-sourced, English-marketed D2C olive oil brand that surfaces at approximately 14 percent matched-subset Presence — is added to the Spanish cohort, the aggregate rises to 14.2 percentage points and breaches the 12.5-percent threshold. The 2.7-point swing is entirely Graza. Graza is the cleanest available diagnostic for the brand-marketing-language tier: a brand that sources its olives from Spain but markets to American consumers in English, indexes in English-language retail feeds, and is discussed in English-language food media. Including Graza in the Spanish cohort breaches the H5 threshold; excluding Graza confirms it. The brand-marketing-language tier — which v0.6 §4.4 identified — is the binding variable. The v0.9 contribution is making the v0.6 refinement testable at threshold and surfacing Graza as the diagnostic case that isolates it cleanly from country-of-origin.

3.6 Finding 6 — Cross-model relative behavior is stable.

For each of the five categories, the per-brand spread between Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini at t_2 correlated with the same-pair spread at t_1 at Pearson $r \geq 0.7$. The result confirmed at 5 of 5 categories, above the 4-of-5 threshold for headline confirmation. The finding indicates that the cross-model differential — what each model emphasizes versus what its companion emphasizes within a category — is preserved across the seven-day interval. Divergent brand-level shifts between the two models would suggest model-specific behavior change; the observed stability instead suggests that whatever AI behavior changed between t_1 and t_2 affected both models in approximately parallel ways, or affected neither.

3.7 Post-hoc — Pattern 1 cross-model spread replicates.

A post-hoc test of v0.6’s Pattern 1 (Cross-Model Variance and Discourse Coherence) at the category level was performed against the same matched-subset data as H1 through H6. The test is post-hoc rather than pre-registered: the H3 statistic is within-category brand-presence variance, while Pattern 1’s statistic is between-

model spread per brand averaged within a category — a different aggregation. The post-hoc test computes, for each category, the mean absolute $|\text{Sonnet} - \text{gpt-5.4-mini}|$ Presence across registered brands, and ranks categories by this mean cross-model spread at each wave. The Spearman rank correlation between t_1 and t_2 rankings is $\rho = 0.800$; the rank correlation between v0.9's t_2 ranking and v0.6's narrative ordering (personal finance > olive oil > project management > skincare > running) is $\rho = 0.900$. Pattern 1 as v0.6 stated it replicates at v0.9. The t_2 ranking matches v0.6's narrative ordering more closely than t_1 does (ρ 0.900 versus 0.600), with finance cross-model spread increasing (+3.91pp) and project management spread decreasing (−1.43pp) between waves — both within H1's empirical noise floor (\$3.1) but in directions that strengthen v0.6's qualitative ordering at t_2 .

Per-category cross-model spread (matched subset, mean $|\text{Sonnet} - \text{gpt-5.4-mini}|$ in pp)

Category	t_1 spread	t_2 spread	Δ
Personal finance	13.0pp	16.9pp	+3.9pp
Project management	14.7pp	13.3pp	−1.4pp
Olive oil	10.9pp	11.9pp	+0.9pp
Running shoes	8.6pp	7.0pp	−1.6pp
Skincare	6.7pp	7.2pp	+0.5pp

The post-hoc test is reported as exploratory and is not part of the formal H1 through H6 score.

3.8 Exploratory: mode-distribution shifts.

After retroactive mode classification of v0.6 and forward classification of v0.9 (per Decision #2), the mode-share distributions on the matched subset were compared per category. The largest within-category mode shifts include: project management software (brand mode +7.3 percentage points, authority mode −4.1pp), olive oil (brand mode −8.3pp, mixed mode +5.3pp, component mode +5.2pp), and skincare (component mode −6.2pp, mixed mode +4.2pp). Personal finance and running shoes were comparatively stable on mode share. The mode-distribution observation is exploratory rather than pre-registered because no prior calibration existed for what threshold counts as “meaningful shift” in mode share at this stage of the program. The v0.9 distributions are reported as a calibration baseline for a future v0.10 pre-registered mode-stability hypothesis with thresholds derived from observed v0.9 noise floors.

4. Pre-Registration Outcomes

The eight pre-registered outcomes (six formal hypotheses plus the H5 sensitivity check plus the exploratory mode observation) and the post-hoc Pattern 1 replication test are reported in Table 1.

Table 1. v0.9 longitudinal re-baseline pre-registration outcomes plus post-hoc Pattern 1 test.

Hyp.	Claim (matched subset, $t_1 \rightarrow t_2$)	Threshold	Outcome	Band
H1	$\geq 70\%$ within $\pm 5pp$; $\geq 90\%$ within $\pm 10pp$	both	88.3% within $\pm 5pp$; 99.0% within $\pm 10pp$	CONFIRMED
H2	Top-3 at t_1 remain in top-5 at t_2 , per category	5 of 5	5 of 5	CONFIRMED
H3	Spearman $\rho \geq 0.7$ on within-category variance ranking	$\rho \geq 0.7$	$\rho = 0.8$	CONFIRMED
H4	Mint within $\pm 5pp$ gross Presence (stability predicted)	$\pm 5pp$	-3.1pp	STABILITY (predicted)
H5 (strict)	Spanish $\leq 12.5pp$ AND K-beauty $\leq 5pp$	both	11.5pp; 1.0pp	CONFIRMED
H5 (incl. Graza)	as H5 with Graza in Spanish cohort	both	14.2pp; 1.0pp	PARTIALLY CONFIRMED
H6	Pearson $r \geq 0.7$ in $\geq 4/5$ categories	≥ 4 of 5	5 of 5	CONFIRMED
Mode dist.	exploratory; no threshold	—	reported in §3.8	EXPLORATORY
Pattern 1 (post-hoc)	Spearman ρ on cross-model spread ranking, t_1 vs t_2	post-hoc	$\rho = 0.800$ (t_1/t_2); $\rho = 0.900$ ($t_2/v0.6$)	POST-HOC REPLICATED

The headline result — five formal hypotheses confirmed plus the sixth landing at the predicted stability band, with the H5 sensitivity producing a sharper instance of v0.6’s refined Pattern 4 hypothesis and the post-hoc Pattern 1 test replicating v0.6’s qualitative ordering — is a clean longitudinal validity outcome for the AIAS measurement program.

5. Discussion: AI Presence as a Real Underlying Construct

The combined H1 through H6 result set, supplemented by the post-hoc Pattern 1 replication, supports a specific interpretive claim: AI Presence behaves like a real, structured property of the LLM tier rather than like measurement noise. The argument has four legs.

First, drift is small. H1 establishes that 88.3 percent of brand-level deltas across the five categories are within ± 5 percentage points across a seven-day interval, on the matched subset. If AI Presence were a noisy observation of an unstructured underlying signal, the per-brand deltas would distribute more widely and more uniformly across the brand population. The observed concentration near zero is the empirical signature of a stable underlying quantity that the measurement is recovering with bounded noise.

Second, leaderboards are stable. H2 establishes that the top-three brands at t_1 remained in the top-five at t_2 in every category. A noise-dominated process would, with non-trivial probability, reorder leaderboards across an interval; the observed across-the-board stability indicates that the position of dominant brands within a category is not artefactual.

Third, structural variance patterns replicate at two scales. H3 establishes that the within-category brand-presence variance ordering across the five categories preserves between waves at Spearman $\rho = 0.8$. The post-hoc Pattern 1 test (§3.7) establishes that the between-model spread ordering — v0.6’s Pattern 1 statistic — preserves at $\rho = 0.800$ between waves and at $\rho = 0.900$ against v0.6’s qualitative narrative ordering. Both within-category brand variance and between-model spread are stable, and both are predicted by the v0.6 “discourse coherence” framing: fragmented-discourse categories produce both wider within-category brand variance and wider between-model spread, while coherent-discourse categories produce narrower variance on both scales. The joint replication of two related but distinct variance statistics moves Pattern 1 from a single-time-point qualitative observation to a confirmed structural property at the longitudinal time scale.

Fourth, established mechanisms hold. H4 establishes that the phantom-brand persistence mechanism observed for Mint in v0.6 §4.6 holds across the longitudinal interval at gross Presence; Mint stays at near-44-percent matched-subset Presence despite being operationally defunct since 2024. The naive-phantom-rate-versus-caveated-phantom distinction unpacked in v0.7 §3.2 is not measured here and remains future

work; the gross-Presence variable common to v0.6 and v0.7 is what H4 confirms longitudinal stability for. H5 establishes that the discourse-language bias mechanism articulated in v0.6 §4.4 (in its refined brand-marketing-language form) holds at t_2 for olive oil and skincare. H6 establishes that cross-model relative behavior is preserved across the interval. Each of the three mechanisms has independent prior support from v0.6 (and from v0.7, v0.8 follow-ups for two of them); the fact that they jointly hold across a longitudinal interval rather than only at a snapshot is what longitudinal evidence contributes to the construct-validity argument.

The four legs together motivate interpretation of AI Presence as a measurable construct rather than as an instrument-and-occasion artifact. The interpretation is stronger than any single hypothesis taken in isolation; the multi-pattern joint replication is what changes the inferential ceiling.

Two findings beyond the formal hypothesis set deserve discussion. *The H5 strict-versus-inclusive sensitivity surfaces the cleanest available diagnostic for v0.6 §4.4's refined hypothesis.* The v0.6 paper had already articulated the brand-marketing-language framing using the Tatcha-versus-Beauty-of-Joseon diagnostic. v0.9 H5 makes the framing testable at numerical threshold and surfaces Graza as a sharper instance: a brand whose olives come from Spain but whose marketing, retailer indexing, and discourse coverage are entirely English. Including Graza in the Spanish cohort breaches the H5 threshold; excluding Graza confirms the threshold. The brand-marketing-language tier is the binding variable. The contribution of v0.9 is therefore to operationalize, threshold, and longitudinally replicate v0.6's refined hypothesis — not to introduce it.

The K-beauty registry coverage is itself a finding. The v0.6 skincare registry contains thirty-one brands, of which exactly one (Beauty of Joseon) is a Korean-headquartered K-beauty brand discoursed primarily in Korean-language category media. The under-representation in the registry construction parallels the discourse-language bias the test is designed to detect. The implication is that the registry-construction process inherits, at the analyst tier, the same Anglo-discourse coverage gap that produces the LLM-tier discourse-language bias. The observation does not affect the H5 K-beauty score (which uses Beauty of Joseon's measured Presence regardless of registry coverage) but constrains how confidently the K-beauty under-surfacing can be interpreted. If a registry constructed with full K-beauty population coverage produced a different H5 K-beauty aggregate, the interpretation would shift; the v0.9 measurement cannot rule that out. The observation is therefore reported as a methodological reflexivity caveat: the test is partially measuring its own registry construction. The reflexivity is, however, internally consistent with the discourse-language bias hypothesis itself — it would be more surprising if registry construction were free of the bias the registry is designed to detect.

6. Limitations

Seven caveats apply to the findings reported here.

Single-pair longitudinal subset. The longitudinal claim is computed on the matched two-model subset (Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini) — the only models present at both t_1 and t_2 . The four parallel-baseline models added at t_2 contribute parallel new baselines but cannot contribute to the longitudinal claim. The validity argument generalizes most directly to the matched-subset model behavior; whether the four added models would, in their own future longitudinal measurements, exhibit similar drift profiles is an open empirical question.

Seven-day inter-measurement interval. The t_1 to t_2 interval is approximately seven days (29–30 April 2026 to 7 May 2026). The interval is short enough that no major model release or training cutoff change is expected to fall within it, supporting the assumption that drift observed across the interval is incidental rather than structural. Longer intervals (one month, three months, twelve months) are required to characterize drift across model-pipeline transitions, and the v0.9 measurement does not address that timescale.

Gross-Presence-versus-naive-phantom-rate distinction not measured at v0.9. The H4 finding is stability of *gross Presence* — the variable v0.6 §4.6 reported and the variable on which the pre-registration tested. The v0.7 reframe distinguished naive-phantom rate (no caveat) from caveated phantom (model presents brand alongside knowledge of its decommissioning). v0.9 measures the gross-Presence variable common to both framings and confirms its longitudinal stability; whether the naive-phantom rate alone is similarly stable is a separable question that a v0.10 pre-registration could address.

Registry-construction reflexivity. The K-beauty observation in §5 establishes that the registry-construction process inherits Anglo-discourse coverage bias. The reflexivity does not invalidate the H1 through H6 results — which are computed on the registered brand population whatever its construction — but constrains their generalization. Stronger generalization would require a registry constructed with full population coverage of all relevant lineages, which is feasible in some categories (skincare) but not in others (premium olive oil includes thousands of small producers whose enumeration would be infeasible).

Construct validity remains open. The longitudinal validity argument in §5 establishes that AI Presence behaves like a real LLM-tier construct. Whether AI Presence correlates with consumer awareness, purchase intent, or category share at the consumer-facing tier is the separate question the AIAS Phase 3 program will address. The v0.9 result unblocks Phase 3 (which requires three categories at two time points and now has data for five) but does not itself constitute construct-validity evidence at the consumer tier.

Personal finance metadata anomaly. The v0.6 personal finance measurement carries the `brand_registry_version`

stamp v2-skin care, inherited from the prior session's module-level constant. The metadata is cosmetic; raw-response inspection confirms that the prompts and registry that fired at v0.6 were the personal finance prompts and registry. The anomaly is disclosed for transparency. If subsequent investigation reveals any subtle dependence on the registry-version stamp through the analysis pipeline, the personal finance H4 (Mint persistence) result would need re-examination; no such dependence has been identified.

Mode-classification audits deferred. Manual inter-rater agreement audits on the v0.9 mode-classified data have not been completed at the time of this paper's posting. The v0.8 audit established 96-percent strict agreement on the brand-surfacing macro unit (the unit at which H1 through H6 score) and 68-percent strict agreement on the underlying five-mode taxonomy. The v0.9 H1 through H6 results score against the macro unit and are therefore not dependent on the strict-mode-taxonomy agreement budget. The exploratory mode-distribution observation in §3.8 is more directly dependent on classifier reliability and is reported with that caveat in mind. Manual audits are scheduled as a methodology disclosure prior to v0.10.

7. Future Research

The Phase 3 program will address five priorities surfaced by the v0.9 findings.

Construct-validity correlation. The v0.9 result unblocks the Phase 3 construct-validity study by providing five categories at two time points. Candidate consumer-tier validators include search-volume data (Google Trends), retail-tracking data (Amazon Best Seller ranks; category-share trackers where available), DTC brand-tracking instruments, and survey-based brand-awareness measurements. The brand-marketing-language framing predicts that consumer awareness should correlate with AI Presence after controlling for production volume and category prestige. The construct-validity work is the program's gate to v1.0 release.

Longer-interval longitudinal measurements. Drift across one-month, three-month, and twelve-month intervals — particularly across model-pipeline transitions or training-cutoff changes — would characterize the temporal stability of AI Presence at scales that bracket model retraining, brand-discourse evolution, and consumer-behavior cycles. The v0.9 seven-day interval establishes a noise floor; longer intervals would add structural drift on top of the noise floor and reveal which categories and brand-types are most temporally stable.

Cross-lingual category expansion. Premium tea and traditional spirits remain candidate categories for designed-for-test measurements of Pattern 4, both satisfying the four cross-lingual structural criteria stated in v0.8. The v0.9 H5 sensitivity — which surfaces the brand-marketing-language tier as the binding variable via the Graza diagnostic — informs the design of future cross-lingual studies: the discourse-language coverage of each registered brand becomes a primary variable, with country of origin as a secondary

controlled covariate.

Naive-phantom-rate longitudinal stability. Extending the v0.7 reframe into the longitudinal frame — measuring whether Mint’s naive-phantom rate (uncaveated mention rate) is as stable as its gross Presence — would distinguish the recommendation-slot persistence component of phantom-brand surfacing from the cumulative-discourse-mass component. A v0.10 pre-registration on this variable specifically is feasible against the v0.9 dataset (the raw responses are deposited; only the classifier needs to be applied).

Registry-construction protocols for cross-lingual categories. The K-beauty observation suggests a structural improvement to the AIAS registry-construction methodology for cross-lingual categories. A formal protocol that explicitly probes for non-English-discourse coverage gaps (for example, through the requirement that any registry intended to measure Pattern 4 in a category include a stated population-share survey of non-English-discourse producers) would mitigate the reflexivity caveat in §5. The protocol revision is planned as part of AIAS Presence Measurement Protocol v1.2.

References

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623.

Brown, T. B., Mann, B., Ryder, N., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33.

González Castro, P. U. (2026). AI Availability: Extending Mental and Physical Availability into Algorithmic Retrieval. *SSRN Working Paper*. <https://ssrn.com/abstract=6659000>

González Castro, P. U. (2026). AI Presence Measurement Across Consumer Categories: A Cross-Category Baseline of Brand Visibility in Large Language Model Outputs. *SSRN Working Paper*. <https://ssrn.com/abstract=6720959>

González Castro, P. U. (2026). A Designed-for-Test Measurement of Phantom-Brand Presence in Large Language Model Outputs: Pre-Registered Evidence from Bed Bath & Beyond, with Pier 1 as Structural Comparator. *SSRN Working Paper*. <https://ssrn.com/abstract=6721779>

González Castro, P. U. (2026). AIAS Presence Measurement Protocol v1.1. *SSRN Working Paper*. <https://ssrn.com/abstract=6722319>

González Castro, P. U. (2026). A Designed-for-Test Measurement of Discourse-Language Bias in Large Language Model Brand Recommendations: Pre-Registered Evidence from Premium Kitchen

Knives, with Marketing-Language Coverage as the Candidate Mechanism. *SSRN Working Paper*. <https://ssrn.com/abstract=6728000>

International Olive Council. (2025). World olive oil and table olive figures. <https://www.internationaloliveoil.org/>

Keller, K. L. (2013). *Strategic Brand Management: Building, Measuring, and Managing Brand Equity* (4th ed.). Pearson.

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11), 2600–2606.

Romaniuk, J. (2023). *Better Brand Health: Measures and Metrics for a How Brands Grow World*. Oxford University Press.

Romaniuk, J., & Sharp, B. (2016). *How Brands Grow: Part 2 — Including Emerging Markets, Services, Durables, New and Luxury Brands*. Oxford University Press.

Sharp, B. (2010). *How Brands Grow: What Marketers Don't Know*. Oxford University Press.

Declarations

Conflict of Interest

The author serves as Director, Corporate Brand Creative and Governance at Samsung Electronics America. The research presented here is independent of Samsung Electronics America and does not constitute Samsung research. No Samsung Electronics America data, personnel, or commercial interests influenced the design, conduct, analysis, or reporting of this study. Samsung Electronics America did not review the manuscript prior to posting. The brand population evaluated in this study (the v0.6 cross-category baseline of project management software, premium running shoes, premium olive oil, premium facial skincare, and personal finance applications) does not include Samsung product lines.

Funding

Self-funded. No external funding sources contributed to this research.

Data Availability

Underlying datasets, the locked pre-registration document (PRE_REGISTRATION_v09_rebaseline_v1.0.md, locked at git commit f8ceb8d, tag v0.9-prereg-locked, lock date 6 May 2026 prior to t_2 data collection), the methodology decision note (docs/v09_DECISION_NOTE.md, locked at git commit e6428a4 prior to the

pre-registration), the per-category registry files (frozen at v0.6's final state), the canonical scoring tables, the post-hoc Pattern 1 replication script and output, the auxiliary unknowns classification, the cross-lab AI audit samples, and the build-pipeline source code are deposited in an Open Science Framework project at [https://osf.io/\[id-pending\]/](https://osf.io/[id-pending]/). The deposit contains row-level CSV outputs throughout — the v0.9 measurement was conducted within a git-tracked repository from methodology lock through publication, and no provenance gaps obtain. The OSF MANIFEST.md documents the file inventory, the lineage between raw measurements and derived analyses, and the cryptographic git-commit anchors for the methodology decision note, the locked pre-registration, and the published scoring tables.

Author Information

Pablo Ulpiano González Castro is faculty in the MPS Branding Program at the School of Visual Arts, New York, where he teaches Brand Transformation through Human-Centered Methodologies. He maintains Third System™ (research entity; data archive and methodology venue) as the publication venue for the AIAS measurement program. Correspondence: pablou@pablou.com · pablou.com.